

RAG Practice Notes

Sample PDF for Day 9 PDF text-extraction practice.

Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions.

Supervised Learning

Supervised learning uses labelled training data. The model learns a relationship between input features and a known target or output.

Unsupervised Learning

Unsupervised learning works with data without labelled target values. It can be used to discover patterns or groups in data.

Classification

Classification is a machine learning task in which the model predicts a category, such as spam or not spam, pass or fail, or real or fake.

Regression

Regression is a machine learning task in which the model predicts a continuous numerical value, such as salary, house price, or marks.

Embeddings

An embedding is a numerical representation of information that captures semantic meaning. Similar pieces of text can have similar embeddings.

RAG

Retrieval-Augmented Generation combines information retrieval with language generation. Relevant information is retrieved from an external knowledge source and provided to an LLM as context for generating an answer.

Chunking

Chunking is the process of splitting a large document into smaller pieces. These chunks can be converted into embeddings and stored for similarity search.

Cosine Similarity

Cosine similarity is commonly used to compare embedding vectors and estimate how similar their directions are. A higher similarity generally indicates greater semantic similarity.